HTML Complete Reference Guide

Comprehensive HTML Training Manual

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1. Introduction to HTML

What is HTML?

HTML (HyperText Markup Language) is the standard markup language for creating web pages. It provides the structure and content of websites, describing how information should be displayed in a web browser.

Key Characteristics:

- **Markup Language**: Uses tags to define elements
- **Not a Programming Language**: Describes structure, not logic
- **Foundation of Web**: Works with CSS and JavaScript
- **Universal Standard**: Supported by all browsers

How HTML Works

1. **You write** HTML code in a text editor
2. **Browser reads** the HTML file
3. **Browser interprets** the tags
4. **Browser displays** the formatted content

HTML History

- **1991**: HTML created by Tim Berners-Lee
- **1995**: HTML 2.0 released
- **1997**: HTML 4.0 introduced
- **2014**: HTML5 became official standard
- **Present**: HTML5 is the current version

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2. HTML Document Structure

Basic HTML Template

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description" content="Page description for SEO">
    <meta name="keywords" content="html, tutorial, learning">
    <meta name="author" content="Your Name">
    <title>Page Title - Shows in Browser Tab</title>
    <link rel="stylesheet" href="styles.css">
    <link rel="icon" href="favicon.ico" type="image/x-icon">
</head>
<body>
    <!-- Your visible content goes here -->
    <h1>Welcome to My Website</h1>
    <p>This is the main content area.</p>

    <script src="script.js"></script>
</body>
</html>
```

Document Structure Breakdown

1. DOCTYPE Declaration

```
<!DOCTYPE html>
```

- Tells browser this is an HTML5 document
- Must be the first line
- Not case-sensitive but conventionally uppercase

2. HTML Root Element

```
<html lang="en">
```

- Root element containing all content
- `lang` attribute specifies language (important for accessibility and SEO)

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3. Head Section

```
<head>
   <!-- Metadata goes here -->
</head>
```

- Contains metadata about the document
- Not displayed on the page
- Includes: title, links to CSS, meta tags, scripts

4. Meta Tags

```
<meta charset="UTF-8">
```

- Defines character encoding (UTF-8 supports all languages)

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

- Essential for responsive design
- Controls how page displays on mobile devices

```
<meta name="description" content="Brief page description">
```

- Page description for search engines
- Shows in search results

```
<meta name="keywords" content="keyword1, keyword2">
```

- Keywords for SEO (less important now)

5. Title Element

```
<title>My Page Title</title>
```

- Shows in browser tab
- Used by search engines
- Required in every HTML document

6. Body Section

```
<body>
```

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```
    <!-- All visible content here -->
</body>
```

- Contains all visible content
- Everything users see on the page

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3. Text Elements

Headings

HTML provides six levels of headings:

```
<h1>Main Heading - Most Important</h1>
<h2>Secondary Heading</h2>
<h3>Tertiary Heading</h3>
<h4>Fourth Level Heading</h4>
<h5>Fifth Level Heading</h5>
<h6>Sixth Level Heading - Least Important</h6>
```

Best Practices:

- Use only one `<h1>` per page
- Don't skip heading levels (don't jump from h1 to h3)
- Use headings for structure, not styling
- Think of headings as document outline

Paragraphs

```
<p>This is a paragraph. It's the most common text element in HTML.</p>
<p>Each paragraph tag creates a new block of text with spacing.</p>
```

Text Formatting

```
<!-- Bold/Strong -->
<strong>This text is bold and semantically important</strong>
<b>This text is bold but not necessarily important</b>

<!-- Italic/Emphasis -->
<em>This text is italic and emphasized</em>
<i>This text is italic but not emphasized</i>

<!-- Other Formatting -->
<mark>Highlighted text</mark>
<small>Smaller text</small>
<del>Deleted/strikethrough text</del>
<ins>Inserted/underlined text</ins>
<sub>Subscript text</sub>
<sup>Superscript text</sup>
<code>Inline code</code>
```

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```
<kbd>Keyboard input</kbd>
<samp>Sample output</samp>
<var>Variable</var>
```

Semantic vs Non-Semantic

Semantic (Use these):

- `<strong>` - Important text
- `<em>` - Emphasized text

Non-Semantic (Use only for styling):

- `<b>` - Bold text
- `<i>` - Italic text

Line Breaks and Horizontal Rules

```
<!-- Line Break -->
<p>First line<br>Second line<br>Third line</p>

<!-- Horizontal Rule (divider) -->
<p>Section 1</p>
<hr>
<p>Section 2</p>
```

Quotations

```
<!-- Blockquote - for longer quotes -->
<blockquote cite="https://source.com">
    This is a longer quote that will be indented and styled as a block.
</blockquote>

<!-- Inline Quote -->
<p>He said, <q>This is a short quote</q>, and left.</p>

<!-- Citation -->
<p><cite>The Great Gatsby</cite> by F. Scott Fitzgerald</p>

<!-- Abbreviation -->
<p>The <abbr title="World Wide Web">WWW</abbr> was invented in 1989.</p>
```

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Preformatted Text

```
<pre>
   Text in a pre element
   is displayed in a fixed-width font,
   and it preserves
   both      spaces and
   line breaks.
</pre>
```

Address Element

```
<address>
   Written by John Doe<br>
   Email: john@example.com<br>
   123 Main Street<br>
   City, State 12345
</address>
```

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4. Lists

Unordered Lists

```
<ul>
   <li>First item</li>
   <li>Second item</li>
   <li>Third item</li>
</ul>
```

List Style Types:

```
<ul style="list-style-type: disc;">  <!-- Default: filled circles -->
<ul style="list-style-type: circle;"> <!-- Empty circles -->
<ul style="list-style-type: square;"> <!-- Squares -->
<ul style="list-style-type: none;">   <!-- No markers -->
```

Ordered Lists

```
<ol>
   <li>First step</li>
   <li>Second step</li>
   <li>Third step</li>
</ol>
```

Ordered List Attributes:

```
<ol type="1">  <!-- Numbers (default) -->
<ol type="A">  <!-- Uppercase letters -->
<ol type="a">  <!-- Lowercase letters -->
<ol type="I">  <!-- Uppercase Roman numerals -->
<ol type="i">  <!-- Lowercase Roman numerals -->

<ol start="5">  <!-- Start from 5 -->
<ol reversed>   <!-- Reverse order -->
```

Description Lists

```
<dl>
   <dt>HTML</dt>
   <dd>HyperText Markup Language - the standard markup language for web pages.</dd>
```

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```
    <dt>CSS</dt>
    <dd>Cascading Style Sheets - used for styling web pages.</dd>

    <dt>JavaScript</dt>
    <dd>Programming language for web interactivity.</dd>
</dl>
```

Nested Lists

```
<ul>
    <li>Frontend
        <ul>
            <li>HTML</li>
            <li>CSS</li>
            <li>JavaScript</li>
        </ul>
    </li>
    <li>Backend
        <ul>
            <li>Node.js</li>
            <li>Python</li>
            <li>PHP</li>
        </ul>
    </li>
</ul>
```

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5. Links and Navigation

Basic Links

```
<!-- External Link -->
<a href="https://www.example.com">Visit Example</a>

<!-- Internal Link (same website) -->
<a href="about.html">About Page</a>
<a href="/contact.html">Contact (from root)</a>

<!-- Anchor Link (same page) -->
<a href="#section1">Jump to Section 1</a>
<h2 id="section1">Section 1</h2>

<!-- Email Link -->
<a href="mailto:info@example.com">Email Us</a>

<!-- Phone Link -->
<a href="tel:+1234567890">Call Us</a>

<!-- SMS Link -->
<a href="sms:+1234567890">Text Us</a>
```

Link Attributes

```
<!-- Open in New Tab -->
<a href="https://example.com" target="_blank">External Site</a>

<!-- Target Options -->
<a href="url" target="_self">   <!-- Same window (default) -->
<a href="url" target="_blank">  <!-- New window/tab -->
<a href="url" target="_parent"> <!-- Parent frame -->
<a href="url" target="_top">    <!-- Full window -->

<!-- Download Link -->
<a href="document.pdf" download>Download PDF</a>
<a href="image.jpg" download="my-image.jpg">Download with Custom Name</a>

<!-- Security for External Links -->
<a href="https://example.com" target="_blank" rel="noopener noreferrer">
   Safe External Link
</a>
```

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```
<!-- No Follow (SEO) -->
<a href="https://example.com" rel="nofollow">No Follow Link</a>

<!-- Title (Tooltip) -->
<a href="page.html" title="Click to visit page">Hover for tooltip</a>
```

Link Styling with CSS

```
<!-- Add classes for styling -->
<a href="#" class="button">Button-style Link</a>
<a href="#" class="nav-link">Navigation Link</a>
```

Image as Link

```
<a href="page.html">
   <img src="image.jpg" alt="Description">
</a>
```

Navigation Menu Example

```
<nav>
   <ul>
      <li><a href="index.html">Home</a></li>
      <li><a href="about.html">About</a></li>
      <li><a href="services.html">Services</a></li>
      <li><a href="contact.html">Contact</a></li>
   </ul>
</nav>
```

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6. Images and Media

Images

```
<!-- Basic Image -->
<img src="image.jpg" alt="Description of image">

<!-- Image with Dimensions -->
<img src="image.jpg" alt="Description" width="300" height="200">

<!-- Responsive Image (scales with container) -->
<img src="image.jpg" alt="Description" style="max-width: 100%; height: auto;">
```

Image Attributes:

- `src` - Source/path to image (required)
- `alt` - Alternative text (required for accessibility)
- `width` - Width in pixels
- `height` - Height in pixels
- `title` - Tooltip text on hover
- `loading="lazy"` - Lazy loading for performance

Image Paths

```
<!-- Relative Paths -->
<img src="image.jpg">            <!-- Same folder -->
<img src="images/photo.jpg">     <!-- Subfolder -->
<img src="../image.jpg">         <!-- Parent folder -->

<!-- Absolute Path -->
<img src="/images/photo.jpg">     <!-- From root -->

<!-- External URL -->
<img src="https://example.com/image.jpg">
```

Figure and Figcaption

```
<figure>
  <img src="artwork.jpg" alt="Beautiful artwork">
  <figcaption>Figure 1: A beautiful piece of art</figcaption>
</figure>
```

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Picture Element (Responsive Images)

```
<picture>
  <source media="(min-width: 1024px)" srcset="large.jpg">
  <source media="(min-width: 768px)" srcset="medium.jpg">
  <img src="small.jpg" alt="Responsive image">
</picture>
```

Audio

```
<audio controls>
  <source src="audio.mp3" type="audio/mpeg">
  <source src="audio.ogg" type="audio/ogg">
  <source src="audio.wav" type="audio/wav">
  Your browser does not support the audio element.
</audio>
```

Audio Attributes:

- `controls` - Show play/pause controls
- `autoplay` - Start automatically (not recommended)
- `loop` - Repeat when finished
- `muted` - Start muted
- `preload` - How to load: "none", "metadata", "auto"

Video

```
<video width="640" height="360" controls>
  <source src="video.mp4" type="video/mp4">
  <source src="video.webm" type="video/webm">
  <source src="video.ogg" type="video/ogg">
  Your browser does not support the video tag.
</video>
```

Video Attributes:

- `controls` - Show controls
- `autoplay` - Auto-play (use with caution)
- `loop` - Loop video
- `muted` - Start muted
- `poster` - Image to show before playing
- `preload` - Preload option

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Example with Poster:

```
<video controls poster="thumbnail.jpg">
   <source src="video.mp4" type="video/mp4">
</video>
```

Iframe (Embedded Content)

```
<!-- Embed Website -->
<iframe src="https://www.example.com" width="800" height="600"></iframe>

<!-- YouTube Video -->
<iframe width="560" height="315"
   src="https://www.youtube.com/embed/VIDEO_ID"
   frameborder="0"
   allowfullscreen>
</iframe>

<!-- Google Map -->
<iframe src="https://maps.google.com/..." width="600" height="450"></iframe>
```

Iframe Attributes:

- `src` - URL to embed
- `width`, `height` - Dimensions
- `frameborder` - Border (0 for none)
- `allowfullscreen` - Allow fullscreen mode
- `loading="lazy"` - Lazy load iframe

SVG (Scalable Vector Graphics)

```
<!-- Inline SVG -->
<svg width="100" height="100">
   <circle cx="50" cy="50" r="40" fill="red" />
</svg>

<!-- SVG as Image -->
<img src="icon.svg" alt="Icon">
```

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7. Tables

Basic Table Structure

```
<table>
  <tr>
    <th>Header 1</th>
    <th>Header 2</th>
    <th>Header 3</th>
  </tr>
  <tr>
    <td>Data 1</td>
    <td>Data 2</td>
    <td>Data 3</td>
  </tr>
  <tr>
    <td>Data 4</td>
    <td>Data 5</td>
    <td>Data 6</td>
  </tr>
</table>
```

Table Elements:

- `<table>` - Container for table
- `<tr>` - Table row
- `<th>` - Table header cell
- `<td>` - Table data cell

Complete Table Structure

```
<table>
  <caption>Monthly Sales Report</caption>

  <thead>
    <tr>
      <th>Month</th>
      <th>Sales</th>
      <th>Profit</th>
    </tr>
  </thead>

  <tbody>
    <tr>
```

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```
            <td>January</td>
            <td>$10,000</td>
            <td>$2,000</td>
        </tr>
        <tr>
            <td>February</td>
            <td>$15,000</td>
            <td>$3,000</td>
        </tr>
    </tbody>

    <tfoot>
        <tr>
            <td>Total</td>
            <td>$25,000</td>
            <td>$5,000</td>
        </tr>
    </tfoot>
</table>
```

Table Attributes

```
<!-- Column Span -->
<td colspan="2">This cell spans 2 columns</td>

<!-- Row Span -->
<td rowspan="3">This cell spans 3 rows</td>

<!-- Headers for Accessibility -->
<th scope="col">Column Header</th>
<th scope="row">Row Header</th>
```

Complex Table Example

```
<table border="1">
    <caption>Product Comparison</caption>
    <thead>
        <tr>
            <th>Product</th>
            <th>Price</th>
            <th>Features</th>
            <th>Rating</th>
        </tr>
```

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```
  </thead>
  <tbody>
    <tr>
      <td rowspan="2">Product A</td>
      <td>$99</td>
      <td>Feature 1, 2, 3</td>
      <td>4.5/5</td>
    </tr>
    <tr>
      <td colspan="3">Additional information about Product A</td>
    </tr>
    <tr>
      <td>Product B</td>
      <td>$149</td>
      <td>Feature 1, 2, 3, 4</td>
      <td>4.8/5</td>
    </tr>
  </tbody>
</table>
```

Table Styling Tips

```
<!-- Add classes for CSS styling -->
<table class="data-table">
  <tr class="header-row">
    <th class="column-name">Name</th>
    <th class="column-value">Value</th>
  </tr>
</table>
```

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8. Forms

Basic Form Structure

```
<form action="/submit" method="POST">
   <!-- Form elements here -->
   <button type="submit">Submit</button>
</form>
```

Form Attributes:

- `action` - Where to send data
- `method` - How to send (GET or POST)
- `enctype` - Encoding type (for file uploads: "multipart/form-data")
- `autocomplete` - Enable/disable autocomplete
- `novalidate` - Disable HTML5 validation

Input Types

```
<!-- Text Input -->
<input type="text" name="username" placeholder="Enter username">

<!-- Password -->
<input type="password" name="password">

<!-- Email -->
<input type="email" name="email" required>

<!-- Number -->
<input type="number" name="age" min="0" max="120" step="1">

<!-- Date -->
<input type="date" name="birthday">

<!-- Time -->
<input type="time" name="appointment">

<!-- Datetime-local -->
<input type="datetime-local" name="meeting">

<!-- Month -->
<input type="month" name="month">
```

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```
<!-- Week -->
<input type="week" name="week">

<!-- Color -->
<input type="color" name="favorite-color">

<!-- Range (Slider) -->
<input type="range" name="volume" min="0" max="100" value="50">

<!-- File Upload -->
<input type="file" name="document">
<input type="file" name="images" multiple accept="image/*">

<!-- URL -->
<input type="url" name="website" placeholder="https://example.com">

<!-- Tel -->
<input type="tel" name="phone" pattern="[0-9]{3}-[0-9]{3}-[0-9]{4}">

<!-- Search -->
<input type="search" name="query">

<!-- Hidden -->
<input type="hidden" name="user_id" value="12345">
```

Checkboxes and Radio Buttons

```
<!-- Checkboxes (Multiple Selection) -->
<input type="checkbox" id="html" name="skills" value="html">
<label for="html">HTML</label>

<input type="checkbox" id="css" name="skills" value="css">
<label for="css">CSS</label>

<input type="checkbox" id="js" name="skills" value="javascript">
<label for="js">JavaScript</label>

<!-- Radio Buttons (Single Selection) -->
<input type="radio" id="male" name="gender" value="male">
<label for="male">Male</label>

<input type="radio" id="female" name="gender" value="female">
<label for="female">Female</label>
```

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```
<input type="radio" id="other" name="gender" value="other">
<label for="other">Other</label>
```

Labels (Important for Accessibility)

```
<!-- Method 1: Using 'for' attribute -->
<label for="username">Username:</label>
<input type="text" id="username" name="username">

<!-- Method 2: Wrapping input -->
<label>
  Username:
  <input type="text" name="username">
</label>
```

Textarea

```
<textarea name="message" rows="5" cols="50" placeholder="Enter your message">
</textarea>

<!-- With maxlength -->
<textarea name="bio" maxlength="500" rows="4"></textarea>
```

Select Dropdown

```
<!-- Basic Select -->
<select name="country">
  <option value="">Select a country</option>
  <option value="us">United States</option>
  <option value="uk">United Kingdom</option>
  <option value="ca">Canada</option>
</select>

<!-- With Default Selection -->
<select name="size">
  <option value="s">Small</option>
  <option value="m" selected>Medium</option>
  <option value="l">Large</option>
</select>

<!-- Multiple Selection -->
<select name="languages" multiple size="4">
  <option value="html">HTML</option>
```

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```
    <option value="css">CSS</option>
    <option value="js">JavaScript</option>
    <option value="python">Python</option>
</select>

<!-- Grouped Options -->
<select name="car">
    <optgroup label="German Cars">
        <option value="audi">Audi</option>
        <option value="bmw">BMW</option>
    </optgroup>
    <optgroup label="Japanese Cars">
        <option value="toyota">Toyota</option>
        <option value="honda">Honda</option>
    </optgroup>
</select>
```

Buttons

```
<!-- Submit Button -->
<button type="submit">Submit Form</button>
<input type="submit" value="Submit Form">

<!-- Reset Button -->
<button type="reset">Reset Form</button>

<!-- Regular Button (for JavaScript) -->
<button type="button" onclick="doSomething()">Click Me</button>

<!-- Image Button -->
<input type="image" src="submit.png" alt="Submit">
```

Form Validation Attributes

```
<!-- Required Field -->
<input type="text" name="username" required>

<!-- Minimum/Maximum Length -->
<input type="text" name="username" minlength="3" maxlength="20">

<!-- Pattern (Regular Expression) -->
<input type="text" name="zip" pattern="[0-9]{5}" title="5-digit zip code">
```

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```
<!-- Min/Max for Numbers -->
<input type="number" name="age" min="18" max="100">

<!-- Disabled Field -->
<input type="text" name="field" disabled>

<!-- Readonly Field -->
<input type="text" name="field" readonly value="Cannot edit">

<!-- Placeholder -->
<input type="text" name="name" placeholder="Enter your name">

<!-- Autofocus -->
<input type="text" name="search" autofocus>

<!-- Autocomplete -->
<input type="text" name="email" autocomplete="email">
```

Fieldset and Legend

```
<fieldset>
   <legend>Personal Information</legend>

   <label for="fname">First Name:</label>
   <input type="text" id="fname" name="firstname">

   <label for="lname">Last Name:</label>
   <input type="text" id="lname" name="lastname">
</fieldset>
```

Datalist (Autocomplete Suggestions)

```
<input list="browsers" name="browser">
<datalist id="browsers">
   <option value="Chrome">
   <option value="Firefox">
   <option value="Safari">
   <option value="Edge">
</datalist>
```

Complete Form Example

```
<form action="/register" method="POST">
```

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```
    <fieldset>
        <legend>Registration Form</legend>

        <label for="username">Username:</label>
        <input type="text" id="username" name="username" required minlength="3">

        <label for="email">Email:</label>
        <input type="email" id="email" name="email" required>

        <label for="password">Password:</label>
        <input type="password" id="password" name="password" required minlength="8">

        <label for="dob">Date of Birth:</label>
        <input type="date" id="dob" name="dob">

        <label>Gender:</label>
        <input type="radio" id="male" name="gender" value="male">
        <label for="male">Male</label>
        <input type="radio" id="female" name="gender" value="female">
        <label for="female">Female</label>

        <label for="country">Country:</label>
        <select id="country" name="country">
            <option value="">Select Country</option>
            <option value="us">United States</option>
            <option value="uk">United Kingdom</option>
        </select>

        <label for="bio">Bio:</label>
        <textarea id="bio" name="bio" rows="4"></textarea>

        <input type="checkbox" id="agree" name="agree" required>
        <label for="agree">I agree to terms and conditions</label>

        <button type="submit">Register</button>
        <button type="reset">Clear Form</button>
    </fieldset>
</form>
```

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9. Semantic HTML5

Why Use Semantic HTML?

Benefits:

- **SEO**: Search engines understand content better
- **Accessibility**: Screen readers can navigate easily
- **Maintainability**: Code is more readable
- **Future-proof**: Better browser support

Semantic Structure Elements

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Semantic HTML Example</title>
</head>
<body>

   <!-- Header (Site-wide or section) -->
   <header>
      <h1>Website Title</h1>
      <nav>
         <ul>
            <li><a href="#home">Home</a></li>
            <li><a href="#about">About</a></li>
            <li><a href="#contact">Contact</a></li>
         </ul>
      </nav>
   </header>

   <!-- Main Content (Only one per page) -->
   <main>

      <!-- Article (Self-contained content) -->
      <article>
         <header>
            <h2>Article Title</h2>
            <p>Published on <time datetime="2025-01-15">January 15, 2025</time></p>
         </header>

         <p>Article content goes here...</p>
```

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```
        <!-- Section within article -->
        <section>
            <h3>Section Heading</h3>
            <p>Section content...</p>
        </section>

        <footer>
            <p>Author: John Doe</p>
        </footer>
    </article>

    <!-- Another article -->
    <article>
        <h2>Another Article</h2>
        <p>Content here...</p>
    </article>

    <!-- Aside (Related content) -->
    <aside>
        <h3>Related Links</h3>
        <ul>
            <li><a href="#">Related Article 1</a></li>
            <li><a href="#">Related Article 2</a></li>
        </ul>
    </aside>

</main>

<!-- Footer (Site-wide or section) -->
<footer>
    <p>&copy; 2025 Company Name. All rights reserved.</p>
    <address>
        Contact us at <a href="mailto:info@example.com">info@example.com</a>
    </address>
</footer>

</body>
</html>
```

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Semantic Elements Reference

`<header>`

- Top of page or section
- Can contain logo, navigation, heading
- Can have multiple per page

`<nav>`

- Navigation links
- Major navigation blocks only

`<main>`

- Main content of document
- Only one per page
- Directly inside `<body>`

`<article>`

- Self-contained content
- Could stand alone
- Examples: blog post, news article, comment

`<section>`

- Thematic grouping of content
- Should have a heading
- Use when no other semantic element fits

`<aside>`

- Tangentially related content
- Sidebars, pull quotes, ads

`<footer>`

- Footer of page or section
- Copyright, contact, links

`<figure>` and `<figcaption>`

- Images, diagrams, code listings
- With optional caption

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`<time>`

- Dates and times
- Machine-readable format

`<mark>`

- Highlighted/marked text
- Search results, important info

`<details>` and `<summary>`

- Collapsible content
- Built-in interactivity

Other Semantic Elements

```
<!-- Details/Summary (Accordion) -->
<details>
  <summary>Click to expand</summary>
  <p>Hidden content appears here when expanded.</p>
</details>

<!-- Dialog -->
<dialog open>
  <p>This is a dialog box</p>
  <button>Close</button>
</dialog>

<!-- Progress Bar -->
<progress value="70" max="100">70%</progress>

<!-- Meter -->
<meter value="0.7" min="0" max="1">70%</meter>

<!-- Output -->
<form oninput="result.value=parseInt(a.value)+parseInt(b.value)">
  <input type="range" id="a" value="50"> +
  <input type="number" id="b" value="50">
  = <output name="result" for="a b">100</output>
</form>
```

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Non-Semantic vs Semantic

Non-Semantic (Generic containers):

```
<div class="header">Header content</div>
<div class="navigation">Nav content</div>
<div class="article">Article content</div>
```

Semantic (Meaningful elements):

```
<header>Header content</header>
<nav>Nav content</nav>
<article>Article content</article>
```

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10. HTML Best Practices

1. Document Structure

✅ **DO:**

- Always include DOCTYPE
- Use lowercase for tags and attributes
- Close all tags properly
- Indent nested elements consistently
- Use semantic HTML5 elements

❌ **DON'T:**

- Mix uppercase and lowercase
- Leave tags unclosed
- Skip DOCTYPE declaration

2. Accessibility

✅ **DO:**

- Always include `alt` text for images
- Use proper heading hierarchy (h1 → h2 → h3)
- Include `lang` attribute in html tag
- Use labels for form inputs
- Ensure sufficient color contrast
- Make interactive elements keyboard-accessible

❌ **DON'T:**

- Use images without alt text
- Skip heading levels
- Use div/span for interactive elements
- Rely only on color for information

3. Forms

✅ **DO:**

- Use appropriate input types
- Include labels for all inputs
- Use required attribute for validation
- Provide helpful placeholder text
- Group related fields with fieldset

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❌ **DON'T:**

- Use text input for everything
- Forget to associate labels
- Rely only on JavaScript validation

4. SEO Optimization

✅ **DO:**

- Use one h1 per page
- Include meta description
- Use descriptive page titles
- Create meaningful URLs
- Use semantic markup
- Add structured data when appropriate

❌ **DON'T:**

- Use multiple h1 tags
- Leave title/meta tags empty
- Use generic page titles

5. Performance

✅ **DO:**

- Optimize image sizes
- Use lazy loading for images/iframes
- Minimize HTTP requests
- Put scripts at end of body
- Use external CSS/JS files

❌ **DON'T:**

- Load huge images unnecessarily
- Inline all CSS/JavaScript
- Load resources you don't need

6. Code Quality

✅ **DO:**

- Use meaningful class/id names
- Add comments for complex sections
- Keep code organized and indented

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- Validate your HTML
- Use consistent naming conventions

❌ **DON'T:**

- Use cryptic names (div1, x, temp)
- Write everything on one line
- Ignore validation errors

7. Modern HTML5

✅ **DO:**

- Use semantic elements
- Use HTML5 input types
- Use native HTML5 features
- Take advantage of new attributes

❌ **DON'T:**

- Use deprecated tags (font, center)
- Ignore new HTML5 capabilities
- Use tables for layout

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11. HTML Cheat Sheet

Document Structure

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Page Title</title>
</head>
<body>
   <!-- Content -->
</body>
</html>
```

Text Elements

```
<h1> to <h6>     Headings
<p>              Paragraph
<br>             Line break
<hr>             Horizontal rule
<strong>         Important text (bold)
<em>             Emphasized text (italic)
<mark>           Highlighted text
<small>          Smaller text
<del>            Deleted text
<ins>            Inserted text
<sub>            Subscript
<sup>            Superscript
```

Links

```
<a href="url">Link</a>
<a href="url" target="_blank">New Tab</a>
<a href="#id">Anchor Link</a>
<a href="mailto:email">Email</a>
<a href="tel:phone">Phone</a>
```

Images & Media

```
<img src="path" alt="description">
<audio controls><source src="audio.mp3"></audio>
```

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```
<video controls><source src="video.mp4"></video>
<iframe src="url"></iframe>
```

Lists

```
<ul><li>Item</li></ul>        Unordered list
<ol><li>Item</li></ol>        Ordered list
<dl><dt>Term</dt><dd>Def</dd></dl>  Description list
```

Tables

```
<table>
  <thead><tr><th>Header</th></tr></thead>
  <tbody><tr><td>Data</td></tr></tbody>
</table>
```

Forms

```
<form action="/submit" method="POST">
  <input type="text" name="field">
  <textarea name="message"></textarea>
  <select name="option">
    <option value="1">Option 1</option>
  </select>
  <button type="submit">Submit</button>
</form>
```

Semantic HTML5

```
<header>     Page/section header
<nav>        Navigation
<main>       Main content
<article>    Self-contained content
<section>    Thematic grouping
<aside>      Related content
<footer>     Page/section footer
<figure>     Image with caption
<figcaption>  Caption
<time>       Date/time
```

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Common Attributes

id="unique"	Unique identifier
class="name"	Class name
style="css"	Inline CSS
title="text"	Tooltip
data-*="value"	Custom data
hidden	Hide element

Quick Reference: Input Types

text	Plain text
password	Hidden password
email	Email address
number	Numeric input
date	Date picker
time	Time picker
color	Color picker
range	Slider
file	File upload
checkbox	Checkbox
radio	Radio button
submit	Submit button
reset	Reset button
button	Generic button

Resources for Further Learning

Official Documentation:

- MDN Web Docs: https://developer.mozilla.org
- W3C HTML Spec: https://www.w3.org/TR/html52/
- HTML5 Doctor: http://html5doctor.com

Validation Tools:

- W3C Validator: https://validator.w3.org
- HTML5 Outliner: https://gsnedders.html5.org/outliner/

Practice:

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- freeCodeCamp
- Codecademy
- Frontend Mentor

END OF HTML GUIDE

This guide covers everything you need to master HTML. Practice regularly and refer back to this guide as needed!

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